

3.3 Halogen Compounds

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.3 Halogen Compounds
Difficulty	Easy

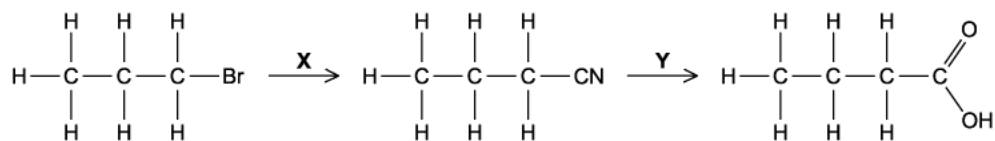
Time allowed: 20

Score: /10

Percentage: /100

Question 1

X and Y are the reagents required to convert 1-bromopropane into butanoic acid.



What are the correct identities of X and Y?

	X	Y
A	NH ₃	HCl(aq)
B	KCN in C ₂ H ₅ OH	NaOH(aq)
C	KCN in C ₂ H ₅ OH	HCl(aq)
D	HCN	NaOH(aq)

[1 mark]

Question 2

Which reaction occurs when ethane and chlorine are mixed in diffused sunlight?

- A. A free-radical substitution with hydrogen given off.
- B. A free-radical substitution with hydrogen chloride given off.
- C. A free-radical substitution with no gas given off.
- D. A nucleophilic substitution with hydrogen chloride given off.

[1 mark]

Question 3

Chlorofluoroalkanes, CFCs, can be used as refrigerants, aerosol propellants and fire extinguishers.

CFCs such as CCl_3F and CCl_2F_2 are more stable than chloroalkanes such as CCl_4 .

What is the reason for their greater stability?

- A. Fluorine has a higher first ionisation energy than chlorine.
- B. Fluorine radicals are more stable than chlorine radicals.
- C. The C–F bond energy is larger than the C–Cl bond energy.
- D. The C–F bond is more polar than the C–Cl bond.

[1 mark]

Question 4

Light initiates the following reaction.



What happens to chlorine in this photochemical reaction?

- A. Heterolytic fission to give an electrophile.
- B. Homolytic fission to give an electrophile.
- C. Heterolytic fission to give a free radical.
- D. Homolytic fission to give a free radical.

[1 mark]

Question 5

High-energy irradiation in the stratosphere produces radicals from chlorofluoroalkanes, commonly known as CFCs.

Which radical could result from this irradiation of $\text{CHFCl/CF}_2\text{Cl}$?

- A. $\text{CHFCl/C}\cdot\text{FCl}$
- B. $\cdot\text{CHCl/CF}_2\text{Cl}$
- C. $\cdot\text{CHFCl/CF}_2\text{Cl}$
- D. $\cdot\text{CFC/CF}_2\text{Cl}$

[1 mark]

Question 6

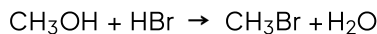
Which reaction is an example of nucleophilic substitution?

- A. $\text{CH}_3\text{CH}_2\text{Br} \rightarrow \text{CH}_2=\text{CH}_2 + \text{HBr}$
- B. $\text{CH}_2=\text{CH}_2 + \text{HBr} \rightarrow \text{CH}_3\text{CH}_2\text{Br}$
- C. $\text{C}_3\text{H}_7\text{Br} + \text{H}_2\text{O} \rightarrow \text{C}_3\text{H}_7\text{OH} + \text{HBr}$
- D. $\text{C}_2\text{H}_6 + \text{Br}_2 \rightarrow \text{C}_2\text{H}_5\text{Br} + \text{HBr}$

[1 mark]

Question 7

Bromomethane, CH_3Br , is used as a fumigant to destroy insect pests in grain that is to be stored. It can be made by reacting methanol with hydrogen bromide.



What type of reaction is this?

- A. Condensation
- B. Electrophilic substitution
- C. Free radical substitution
- D. Nucleophilic substitution

[1 mark]

Question 8

Dichlorodifluoromethane, CCl_2F_2 , has been used in aerosol propellants and as a refrigerant.

Which statement helps to explain why dichlorodifluoromethane is chemically inert?

- A. The carbon-fluorine bond energy is large.
- B. The carbon-fluorine bond has a low polarity.
- C. Fluorine is highly electronegative.
- D. Fluorine compounds are non-flammable.

[1 mark]

Question 9

The reaction of chlorine with methane is carried out in the presence of light.

What is the function of the light?

- A. To break the C–H bonds in methane.
- B. To break up the chlorine molecules into radicals.
- C. To break up the chlorine molecules into ions.
- D. To heat up the mixture.

[1 mark]

Question 10

Which reagents and conditions will convert propane into 1-chloropropane?

- 1 Cl_2 and sunlight
- 2 Conc. HCl , reflux
- 3 PCl_5

- A. 1, 2 and 3
- B. 1 and 2
- C. 2 and 3
- D. 1 only

[1 mark]

Model Answers: Easy

1

The correct answer is **C** because:

- If a halogenoalkane is heated under reflux with a solution of potassium cyanide in ethanol, the halogen is replaced by a -CN group and a nitrile is produced.
 - So, reagent X must be KCN in $\text{C}_2\text{H}_5\text{OH}$.
- Next, to produce a carboxylic acid, the nitrile is heated under reflux with dilute hydrochloric acid.
 - So, reagent Y must be HCl .

A is incorrect as the reaction of a halogenoalkane and ammonia would produce a primary amine.

B & D are incorrect as reagent Y could not be NaOH as this would produce sodium butanoate -COONa instead of -COOH .

2

The correct answer is **B** because:

- Ethane reacts with chlorine by free radical substitution in the presence of sunlight.
- In the presence of sunlight, Cl_2 breaks down to form two chlorine radicals.
 - $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$
- The generated chlorine radical can then react with ethane to give an ethane radical ($\text{C}_2\text{H}_5\cdot$) and HCl .
 - $\text{C}_2\text{H}_6 + \text{Cl}\cdot \rightarrow \text{C}_2\text{H}_5\cdot + \text{HCl}$
- Therefore, the reaction is a free-radical substitution with hydrogen chloride given off.

3

The correct answer is **C** because:

- CFCs and chloroalkanes are similar as they both contain C–C/bonds, but they differ in that CFCs contain C–F bonds.
- The bond energy of the C–F bond is larger than the C–C/bond.
- So, it takes less energy to release C/atoms than F atoms from CFCs.
 - This is why CFCs are more stable than chloroalkanes.

A & B are incorrect as the stability of a molecule comes from the strength of the bonds within the molecule, not the ionisation energy or stability of radicals.

D is incorrect as the polarity of a bond tells us how electronegative an atom is which contributes to the strength of the bond, but it not the direct reason for the stability of the molecule overall.

4

The correct answer is **D** because:

- Homolytic fission = splitting a bond to produce two particles which are the same in the sense that they both have a single unpaired electron (both are free radicals).
- Heterolytic fission = producing two particles which are different because one is a positive ion and the other a negative ion.
- In this photochemical reaction, energy from the sunlight causes chlorine molecules to split into two chlorine atoms.
 - Since chlorine atoms have single unpaired electrons, we call them chlorine free radicals – or more usually just chlorine radicals.
- Free radicals are formed if a bond splits evenly – each atom getting one of the two electrons.
 - Hence, chlorine undergoes homolytic fission to give free radicals.

5

The correct answer is **C** because:

- In the given CFC above, there are:

- C-H bonds, C-F bonds, C-C/bonds and a C-C bond.
- In all the options the C-C bond remains intact.
- The order of bond strength of the remaining 3 bonds is:
 - $C-H > C-F > C-C/$
- Therefore, a C-C/bond will be broken since it requires the least energy - this only occurs in option **C**.

A is incorrect as the CFC loses a F atom, suggesting that a C-F bond is broken.

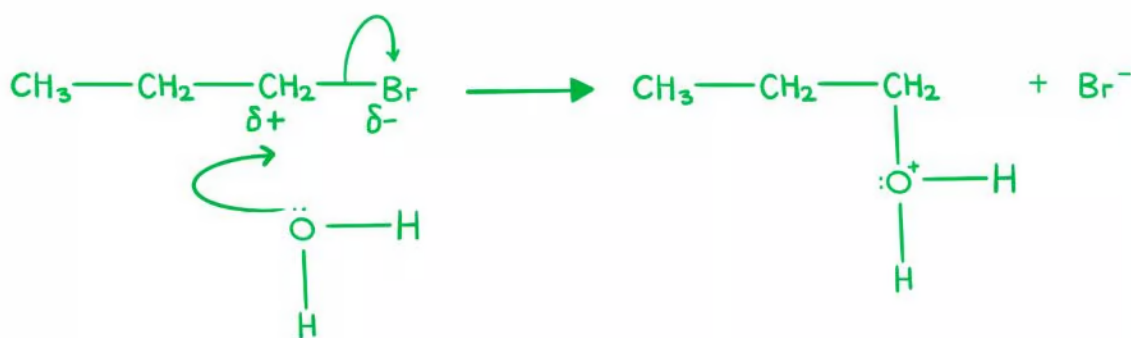
B is incorrect as the CFC loses a different F atom, yet still suggesting that a C-F bond is broken.

D is incorrect as the CFC loses a H atom, suggesting that a C-H bond is broken.

6

The correct answer is **C** because:

- A nucleophile is either a negative charge or a functional group with a lone pair of electrons e.g. -OH
- Nucleophilic substitution reactions involve substitution of a group or atom by a nucleophile.
- The only option which contains a nucleophile reactant is the -OH group from the water molecule in **C**.
- The mechanism for the reaction with bromopropane is as follows:



- Therefore, a C–C/bond will be broken since it requires the least energy – this only occurs in option **C**.

A is incorrect as the CFC loses a F atom, suggesting that a C–F bond is broken.

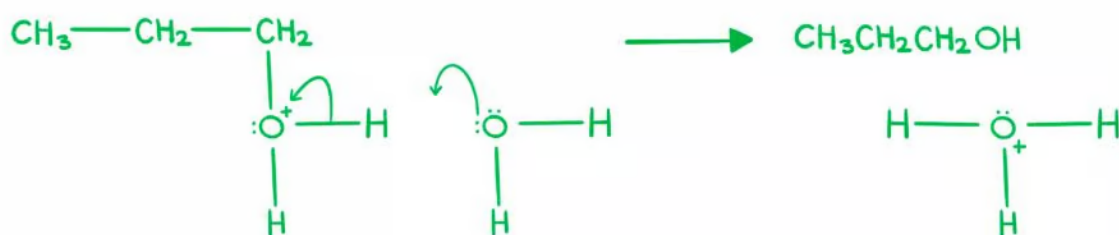
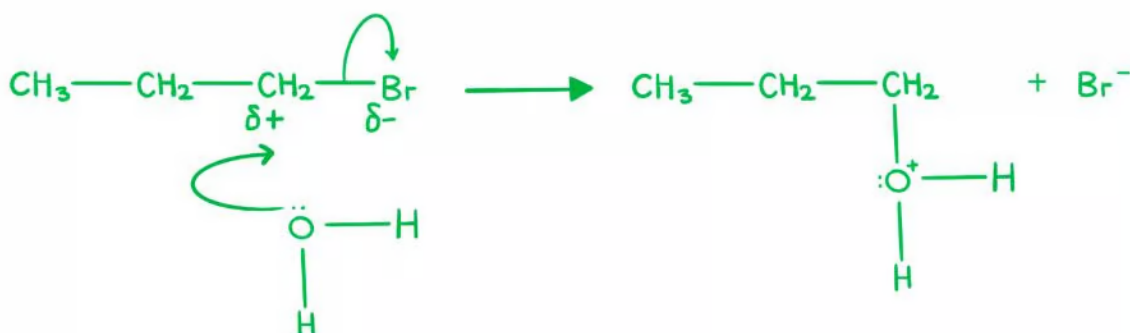
B is incorrect as the CFC loses a different F atom, yet still suggesting that a C–F bond is broken.

D is incorrect as the CFC loses a H atom, suggesting that a C–H bond is broken.

6

The correct answer is **C** because:

- A nucleophile is either a negative charge or a functional group with a lone pair of electrons e.g. –OH
- Nucleophilic substitution reactions involve substitution of a group or atom by a nucleophile.
- The only option which contains a nucleophile reactant is the –OH group from the water molecule in **C**.
- The mechanism for the reaction with bromopropane is as follows:



A is incorrect as this is an elimination reaction.

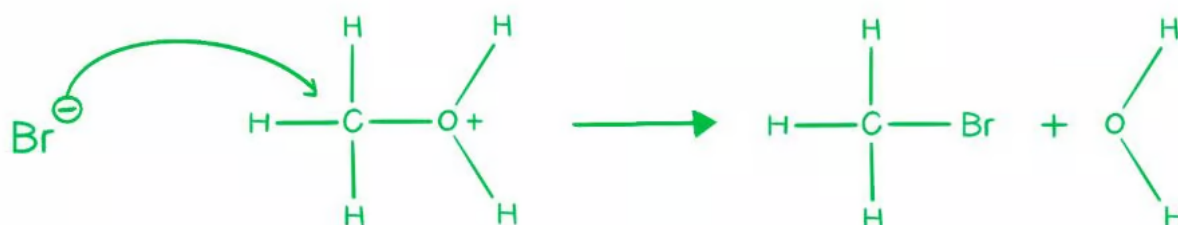
B is incorrect as this is a electrophilic addition reaction.

D is incorrect as this is a free radical substitution.

7

The correct answer is **D** because:

- A nucleophile is either a negative charge or a functional group with a lone pair of electrons e.g. -OH
- Nucleophilic substitution reactions involve substitution of a group or atom by a nucleophile.



8

The correct answer is **A** because:

- In dichlorodifluoromethane, there are:
 - Two C-F bonds and two C-Cl bonds
- Both of these are incredibly strong bonds, and the carbon-fluorine bond energy is the larger of the two.
- The C-F bond strength is the reason for the inertness of CCl_2F_2 and CFCs in

general.

- They are so strong that such bonds don't break under the energy of ordinary visible light.
 - UV light is required to break them.

B is incorrect as the polarity of a bond tells us how electronegative an atom is, which contributes to the strength of the bond, but is not the direct reason for the inertness of the molecule overall.

C is incorrect as halogens being highly electronegative strengthens the bonds due to bond polarity, but is not directly the reason for the inertness.

D is incorrect as this is another consequence of the inertness of CCl_2F_2 , not a reason for it.

9

The correct answer is **B** because:

- This reaction is known as chlorination of methane and proceeds through the radical chain mechanism.
- The radical chain mechanism is characterised by three steps: **initiation**, **propagation** and **termination**.
- Initiation requires an input of energy but after that the reaction is self-sustaining.
 - This energy comes from a high intensity light source, such as the sun.
- Initiation breaks the bond between the chlorine molecules causing them to break up into chlorine radicals.

10

The correct answer is **D** because:

- This reaction is known as chlorination of propane and proceeds through a free radical chain mechanism.
- Initiation requires an input of energy but after that the reaction is self-sustaining.
 - This energy comes from a high intensity light source, such as the sun.

- So, all propane required to convert into 1-chloropropane is chlorine gas (Cl_2) and sunlight.
- The overall reaction is:
 - $\text{CH}_3\text{CH}_2\text{CH}_3 + \text{Cl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{HCl}$

